

Travaya — AI Chatbot Module (Triva)

Module: AI & Recommendation System

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1. File Structure

```
travaya_chatbot/  
|  
|-- chatbot_api.py          <- Flask API server (run this)  
|-- intents.json           <- Model 1: General travel intents  
|-- intents1.json          <- Model 2: India destination-specific  
    details  
|-- TrivaChatbot.jsx        <- React component for MERN frontend  
|  
|-- data.pickle             <- Auto-generated: preprocessed Model 1 data  
|-- data1.pickle            <- Auto-generated: preprocessed Model 2 data  
|-- model_weights.h5        <- Auto-generated: trained Model 1 weights  
|-- model1_weights.h5       <- Auto-generated: trained Model 2 weights  
|  
\\-- README.md              <- Documentation
```

2. How to Run

2.1. Step 1: Install Dependencies

```
pip install flask flask-cors tensorflow nltk
```

2.2. Step 2: Start the Flask Chatbot API

```
python chatbot_api.py
```

2.3. Step 3: Integrate with React (MERN Frontend)

Copy `TrivaChatbot.jsx` into your React project's components folder, then add it anywhere in your app:

```
// In App.jsx or any layout component  
import TrivaChatbot from './components/TrivaChatbot';  
  
function App() {  
  return (  
    <div>  
      { /* Your existing MERN app */ }  
      <TrivaChatbot /> { /* Add this one line */ }  
    </div>  
  );  
}
```

Listing 1: App.jsx — Adding Triva to MERN frontend

The chatbot will appear as a floating button (bottom-right corner) on every page of your website.

3. How the Chatbot Works

3.1. Two-Model Architecture

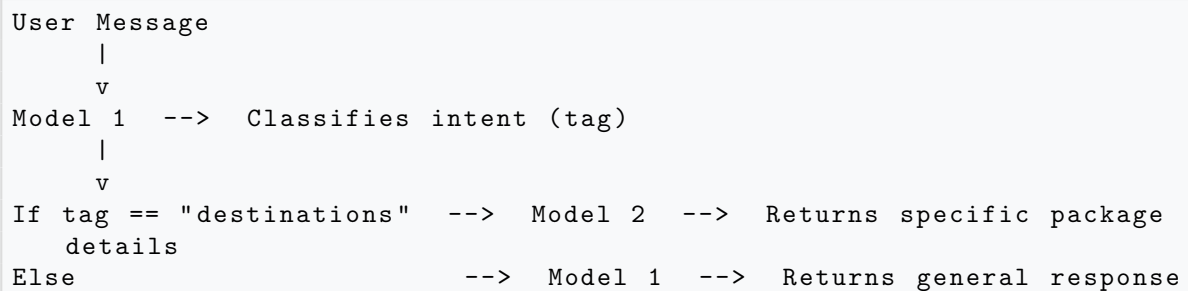
Model 1 — General Intent Classifier

- File: `intents.json`
- Handles: greetings, goodbyes, general package info, booking info, travel styles
- Tags: `greeting`, `goodbye`, `identity`, `destinations`, `budget_travel`, `adventure_travel`, etc.

Model 2 — Destination-Specific Responder

- File: `intents1.json`
- Handles: specific Indian state/destination details with full package info
- Tags: `Goa`, `Kerala`, `Rajasthan`, `Kashmir`, `Andaman`, `Himachal`, `Ladakh`, etc.

3.2. Routing Flow



4. Covered Indian Destinations

Destination	Package Cost	Duration	Type
Goa	INR 12,000 INR 18,000	– 5D/4N	Beach, Water Sports
Andaman & Nicobar	INR 28,000 INR 45,000	– 7D/6N	Beach, Scuba Diving
Kerala	INR 20,000 INR 32,000	– 7D/6N	Backwaters, Nature
Rajasthan	INR 20,000 INR 38,000	– 10D/9N	Heritage, Culture
Kashmir	INR 25,000 INR 40,000	– 6D/5N	Nature, Adventure

Destination	Package Cost	Duration	Type
Himachal Pradesh	INR 22,000 INR 35,000	– 8D/7N	Adventure, Trekking
Ladakh	INR 25,000 INR 42,000	– 7D/6N	Adventure, Biking
Darjeeling & Sikkim	INR 18,000 INR 28,000	– 7D/6N	Nature, Culture
Varanasi & Agra	INR 12,000 INR 20,000	– 5D/4N	Spiritual, Heritage
Northeast (Meghalaya/Assam)	INR 22,000 INR 36,000	– 8D/7N	Nature, Wildlife
Karnataka	INR 14,000 INR 24,000	– 6D/5N	Heritage, Nature
Tamil Nadu	INR 15,000 INR 26,000	– 7D/6N	Culture, Spiritual
Maharashtra	INR 14,000 INR 22,000	– 6D/5N	Heritage, City
Gujarat	INR 16,000 INR 28,000	– 7D/6N	Wildlife, Heritage
Lakshadweep	INR 30,000 INR 55,000	– 5D/4N	Luxury, Beach
Odisha	INR 11,000 INR 18,000	– 5D/4N	Spiritual, Beach

5. API Reference

5.1. POST /api/chat

```
// Request Body
{ "message": "Tell me about Kerala" }

// Response
{
  "status":      "success",
  "response":    "Kerala - God's Own Country\n- Duration: 7 Days...",
  "tag":         "destinations",
  "confidence":  0.94
}
```

Listing 2: Request and Response — /api/chat

5.2. GET /api/health

Returns server status and a list of all supported destinations.

5.3. GET /api/destinations

Returns all destination tags currently supported by the chatbot.

6. Technology Stack

Component	Technology
NLP Engine	TensorFlow 2.x + Keras
Text Processing	NLTK + Lancaster Stemmer
API Server	Flask + Flask-CORS
Frontend Component	React.js (JSX)
Data Format	JSON (intents) + Pickle (preprocessed)
Algorithm	Bag-of-Words + Dense Neural Network

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